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## RESEARCH ARTICLE

# A Systematic Literature Review on System Dynamics

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**ABSTRACT** System Dynamics is a methodology for modeling and simulating dynamically complex systems, which are characterized by feedback loops, nonlinearity, and time delays. It uses Causal Loop Diagrams and Stock-and-Flow Diagrams to represent system structures and to analyze how these structures influence system behavior over time. The majority of reviews on System Dynamics use cases either focus on a single thematic domain or are not up to date. As a result, a comprehensive and up-to-date overview of System Dynamics use cases across all thematic domains is lacking. Furthermore, no Systematic Literature Review has yet examined whether and to what extent artificial intelligence has been employed to extend and enhance the System Dynamics approach in identified use cases. To address this research gap, we conducted a Systematic Literature Review. The identified studies are classified within a sociotechnical system framework consisting of User-Machine Interaction, Value Chain-related Domains, and Social Factors. Additionally, we examine the role of artificial intelligence in advancing the System Dynamics approach within the identified use cases. We identified 41 relevant publications covering 19 thematic domains. Most studies fall within the Value Chain-related Domains system, followed by Social Factors and User-Machine Interaction. The integration of artificial intelligence into System Dynamics use cases remains rare. Our findings highlight the broad applicability of System Dynamics across various thematic domains while simultaneously revealing a gap in the complementary use of artificial intelligence.

**INDEX TERMS** System dynamics, systematic literature review, artificial intelligence, human-machine interaction.

## I. INTRODUCTION

In an increasingly complex and dynamic world, decision-makers face the challenge of understanding and managing interconnected systems. The growing complexity of social, economic, and technical systems often makes it difficult to calculate the full impact of decisions or predict long-term consequences. General Systems Theory, as introduced by Bertalanffy [1], provides a theoretical foundation for systems thinking and enables the analysis of structural relationships and interdependencies in complex systems. A central aspect of this theory is the interaction between systems and their

environment, as well as the various ways in which systems can be organized, ranging from hierarchical to networked and emergent structures [1]. In contrast, the System Dynamics approach, developed by Forrester [2], offers a methodology for modeling and scenario-based simulation of the nonlinear behavior of dynamically complex systems. This is achieved by capturing dynamic interactions through stocks, flows, time delays, and internal feedback loops within the defined system boundaries [2]. Both approaches share a focus on understanding interdependencies and system-level effects, making them highly complementary in addressing complex challenges [3]. Thus, they can be effectively combined to support decision-making in complex problem environments [3]. In recent decades, the application of System

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Dynamics has expanded significantly, encompassing a broad spectrum of disciplines. Several studies, including [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], have conducted reviews on specific thematic domains of System Dynamics use cases. However, a cross-domain synthesis is lacking. Apart from the work of Zanker et al. [4], no Systematic Literature Review (SLR) has examined System Dynamics use cases across all thematic domains. Moreover, no up-to-date SLR extends beyond 2019 while simultaneously covering all thematic domains, leading to an incomplete and outdated understanding of System Dynamics use cases. Existing reviews remain fragmented and do not provide a unified classification that illustrates how System Dynamics use cases have evolved over time and across disciplines. To address this gap, this study reviews existing System Dynamics use cases. Building upon the identified research gap, this study provides the following key contributions:

- 1) We provide a comprehensive and up-to-date overview of System Dynamics use cases.
- 2) We identify a research gap in the complementary use of AI to enhance the System Dynamics approach.

By providing a comprehensive overview of System Dynamics use cases, this SLR contributes to a better understanding of the role of System Dynamics in addressing complex challenges in decision-making. Additionally, this SLR highlights opportunities for integrating AI into System Dynamics methodologies, offering insights for researchers and practitioners seeking to enhance the accessibility and effectiveness of System Dynamics models.

This work is structured as follows: In section II, System Dynamics fundamentals are elaborated to provide a brief introduction to the topic, which is followed by related work. Next, section III describes our SLR methodology. The results from our SLR are then presented in section IV. The results are consequently discussed in section V, followed by a discussion of the implications. Finally, we conclude in section VI. This section is followed by a discussion of the study's limitations, and an outlook on future research.

## II. SYSTEM DYNAMICS FUNDAMENTALS AND RELATED WORK

### A. SYSTEM DYNAMICS FUNDAMENTALS

System Dynamics is a modeling approach designed to analyze and simulate the behavior of dynamically complex systems over time [2]. It captures the structure and behavior of systems through the interaction of stocks, flows, time delays, and internal feedback loops [2]. The methodology relies on two core modeling tools: Causal Loop Diagrams (CLDs) and Stock-and-Flow Diagrams (SFDs) [2]. CLDs illustrate the feedback structures that underlie and explain system behavior, while SFDs represent the quantitative relationships among stocks, inflows, and outflows [2]. System Dynamics models incorporate both reinforcing and balancing feedback loops, as well as time delays that

influence the system response over time [2]. These models are typically implemented in computer-based simulations to analyze long-term system behavior and test policy interventions [2]. System Dynamics was conceptualized in the 1950s by Forrester at the Massachusetts Institute of Technology (MIT) [19]. Initially developed to support corporate and industrial decision-making, the methodology was formally introduced through *Industrial Dynamics* [2], which laid the foundation for analyzing corporate systems using stocks, flows, and feedback loops. Subsequent publications such as *Urban Dynamics* [20] and *World Dynamics* [21] extended its application to social and socioeconomic systems. A pivotal moment came with *The Limits to Growth* [22], commissioned by the Club of Rome. This study used System Dynamics to simulate long-term global sustainability scenarios, significantly raising public and academic awareness of global systemic interdependencies. Since its inception, the System Dynamics approach has been applied across a wide range of thematic domains. Textbooks such as Bossel's *System Zoo* series [23], [24], [25] illustrate how diverse systems, from technical and ecological to economic and social, can be modeled using System Dynamics. In addition to these developments, hybrid approaches have emerged to extend the modeling capabilities of the System Dynamics approach [26]. Furthermore, recent advances in artificial intelligence, particularly the integration of Large Language Models (LLMs), open up new avenues for AI-driven extensions of the methodology [27], [28], [29].

### B. RELATED WORK

This section outlines existing reviews on the use of System Dynamics across different application domains. Tables 1 and 2 summarize selected references that review System Dynamics use cases. A notable example is the study by Zanker et al. [4], which analyzes the general thematic domains of System Dynamics use cases. They conducted a SLR to synthesize the general application areas of System Dynamics from studies published between 2016 and 2019. In addition to this SLR, which explored the general application areas of the System Dynamics approach, several other studies have focused on specific thematic domains. One such domain is strategic management. Cosenz and Noto [6] conducted a systematic analysis to identify application areas of System Dynamics in strategic management. Their study identified nine theoretical perspectives on System Dynamics applications, which were categorized into six thematic domains using a problem-related framework [6]. Their findings indicate that System Dynamics is primarily used as a systemic analysis tool for decision support in this context [6]. Another thematic domain focuses on sustainability-related challenges. Rebs et al. [7] conducted an SLR on System Dynamics applications in sustainable supply chain management. Their study applied a systems thinking perspective to classify the findings into three system classes [7]. Furthermore, their findings highlight that System Dynamics can be

**TABLE 1.** Overview of recent literature reviews regarding System Dynamics. The table displays each review's authors, publication year, and the number of papers included in the review. Additionally, a summary is given, and the applied review structure is provided (Part 1 of 2).

| Authors                       | Year | # of papers included | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Structure applied to categorize literature                                                                                                                                                       |
|-------------------------------|------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zanker et al. [4]             | 2021 | 212                  | The authors conducted a Systematic Literature Review to identify and categorize the current use cases of the System Dynamics methodology. The review analyzed publications from 2016 to 2019, adhering to the PRISMA guidelines. The results were classified using a problem-related framework. They employed quantitative network analysis (VOSviewer) and content analysis (KH Coder). They identified 20 distinct thematic domains. The findings indicate that System Dynamics is frequently complemented by other modeling approaches.                                                                                                        | (1) Business<br>(2) Environment<br>(3) Health<br>(4) Miscellaneous                                                                                                                               |
| Darabi and Hosseinichimeh [5] | 2020 | 301                  | The authors conducted a systematic review of System Dynamics applications in health and medicine. They analyzed publications from 1960 to 2018, categorizing 21 application areas into three main groups using a problem-related approach. The study focused on disease-related modeling, distinguishing between infectious and noninfectious diseases. It found that all infectious disease models were population-based. The review also identified differences in feedback intensity and model transparency, highlighting deficiencies in model formulation disclosure and validation.                                                         | (1) Disease-related Modeling<br>(2) Organizational Modeling<br>(3) Regional Health Modeling                                                                                                      |
| Cosenz and Noto [6]           | 2016 | 172                  | The authors conducted a systematic review of theoretical and empirical literature to explore the development of System Dynamics in strategic management research. The review analyzed publications from 1979 to 2014. The results were classified using a problem-related framework. The study identifies key research themes and developmental patterns, emphasizing the combination of System Dynamics modeling with strategic management applications.                                                                                                                                                                                         | (1) Performance Management<br>(2) Stakeholder Theory and Knowledge Elicitation<br>(3) Strategic Planning<br>(4) Organizational Learning<br>(5) Project Management<br>(6) Supply Chain Management |
| Rebs et al. [7]               | 2019 | 102                  | The authors conducted a systematic review of System Dynamics models in Sustainable Supply Chain Management (SSCM). The study systematizes SSCM from a systems thinking perspective using a conceptual framework and analyzes System Dynamics models for forward, reverse, and closed-loop supply chains. The review analyzed papers published between 1998 and 2017. The review highlights the underrepresentation of System Dynamics modeling in SSCM, identifies key sustainability-related constructs, and discusses the need for hybrid modeling approaches integrating simulation, optimization, and multi-criteria decision-making methods. | (1) Ecological System<br>(2) Social System<br>(3) Economic System                                                                                                                                |

complemented by other methodologies, such as agent-based modeling, discrete-event simulation, life cycle assessment, and multi-criteria decision-making [7]. System Dynamics has also been used in healthcare and medicine. Darabi and

Hosseinichimeh [5] conducted a systematic review of System Dynamics applications in healthcare. Additionally, the study examined common model structures, identifying differences in feedback intensity and model transparency [5]. The

**TABLE 2.** Overview of recent literature reviews regarding System Dynamics. The table displays each review's authors, publication year, and the number of papers included in the review. Additionally, a summary is given, and the applied review structure is provided (Part 2 of 2; continued from Table 1).

| Authors              | Year | # of papers included | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Structure applied to categorize literature                                                                                                                                                                                                                                              |
|----------------------|------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Leopold [10]         | 2016 | 69                   | The author conducted a literature review focusing on System Dynamics models applied within the energy sector. This review systematically analyzed relevant System Dynamics studies published from 2000 to 2015, categorizing them into four main energy-related topics. The Findings indicate that System Dynamics is extensively utilized globally as a decision-support tool in energy policy formulation and planning, predominantly in scenarios involving long-term forecasting and resource management. The review highlights the multidisciplinary nature and widespread adoption of System Dynamics for addressing complex dynamic problems within the energy domain.                                                | <ol style="list-style-type: none"> <li>(1) Fossil fuels</li> <li>(2) Renewable</li> <li>(3) Electricity</li> <li>(4) Further energy-related resources</li> </ol>                                                                                                                        |
| Sedarati et al. [11] | 2019 | 27                   | The authors conducted a Systematic Literature Review to identify and categorize the applications of the System Dynamics methodology within tourism planning and development. The review analyzed peer-reviewed journal publications from 1961 to 2015, across three bibliographic databases. The results were structured into six thematic categories. They applied qualitative content analysis to synthesize the findings. The findings indicate that while System Dynamics shows considerable potential for addressing complex tourism-related issues, its application remains limited and predominantly qualitative.                                                                                                     | <ol style="list-style-type: none"> <li>(1) Multisector applications</li> <li>(2) Attractions sector</li> <li>(3) Adventure and outdoor recreation sector</li> <li>(4) Transportation sector</li> <li>(5) Accommodation sector</li> <li>(6) Event sector</li> </ol>                      |
| Ghisolfi et al. [17] | 2022 | 50                   | The authors conducted a Systematic Literature Review on the System Dynamics methodology in freight transport decarbonization. They reviewed publications from the Google Scholar database up to the year 2021, adhering to the PRISMA guidelines. The results were structured into five decarbonization strategies. They employed qualitative content analysis to synthesize the findings and highlighted relevant external factors and simulated policy instruments. The findings indicate that System Dynamics models currently address a limited combination of decarbonization strategies, emphasizing a significant research gap regarding the explicit modeling of time-dependent policy impacts and system responses. | <ol style="list-style-type: none"> <li>(1) Reducing Freight Transport Demand</li> <li>(2) Shifting Freight to Lower-Carbon Transport Modes</li> <li>(3) Improving Assets Utilization</li> <li>(4) Increasing Energy Efficiency</li> <li>(5) Switching to Lower-Carbon Energy</li> </ol> |
| Saavedra et al. [18] | 2018 | 10                   | The authors conducted a literature review to identify and categorize the use of System Dynamics methodology in sustainable and renewable energy supply chains. They analyzed 10 peer-reviewed journal publications from 2007 to 2017, retrieved from the Scopus database. The results were structured into three analytical categories. The authors employed qualitative content analysis to synthesize the findings. Results indicate that System Dynamics is primarily used in bioenergy contexts, highlighting significant potential but limited current adoption in renewable energy supply chain research.                                                                                                              | <ol style="list-style-type: none"> <li>(1) Biomass Scenario Model</li> <li>(2) Sensitivity Analysis</li> <li>(3) Hybrid Modelling</li> </ol>                                                                                                                                            |

authors also highlighted deficiencies in model transparency and validation, particularly due to unpublished formulas and insufficient validation reports [5]. Ghisolfi et al. [17] conducted a SLR to identify applications of the System Dynamics methodology in freight transport decarbonization, structuring their findings into five decarbonization strategies. Their results highlight a significant research gap regarding the explicit modeling of time-dependent policy impacts and system responses [17]. In tourism planning and development, Sedarati et al. [11] conducted a SLR categorizing applications of System Dynamics into six thematic sectors. They conclude that despite its considerable potential, the use of System Dynamics in tourism remains limited and predominantly qualitative [11].

### III. METHODOLOGY

This chapter outlines the methodology used to explore the current state of knowledge on System Dynamics. We conducted a Systematic Literature Review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines by Page et al. [30] to achieve this goal. An SLR systematically gathers, assesses, and structures all available and relevant evidence on a specific topic. Our objective was to survey and structure the existing knowledge, classify it using theoretical frameworks, and identify limitations and research gaps. Besides these tasks, an SLR also aims to provide insights and guidelines for practitioners in this field. Therefore, we defined all existing knowledge as the published literature in specific databases at a particular time to ensure the quality and recency of our study. We structured our methodology into eight steps, derived and adapted from the PRISMA guidelines, to ensure a comprehensive SLR. These steps are illustrated in Figure 1.

Following the guidelines of Page et al. [30], the first step of this Systematic Literature Review defined the research objective as follows: “A Systematic Literature Review on System Dynamics use cases.”

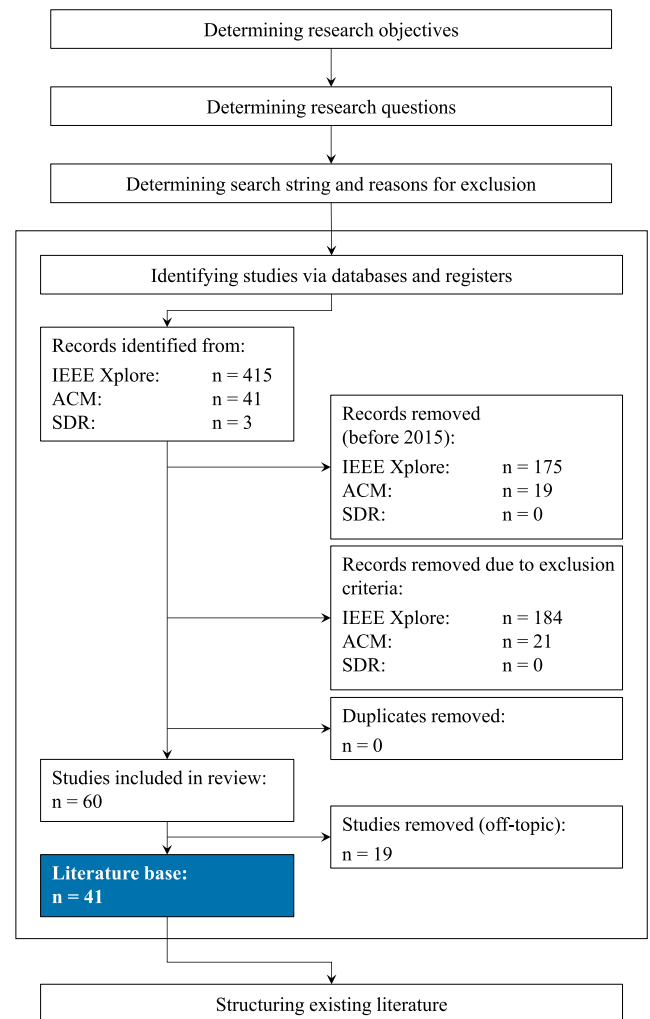
The results are structured according to thematic domains within a systems thinking framework. This approach allows for the categorization of results into distinct thematic domains within systems, providing a clear overview of the methodology’s use cases across various fields.

In the second step, we derived the following research questions from our research objectives:

- What are the use cases of System Dynamics?
- To what extent are AI techniques, particularly LLMs, complementarily applied to the System Dynamics methodology in identified use cases?
- What publication trends can be observed in System Dynamics research over time?

#### A. LITERATURE SEARCH STRATEGY

Following the presented framework, the third step of this SLR was to define the search string and the criteria for excluding individual search results. Papers were included that



**FIGURE 1. Overview of the process for obtaining the literature base as derived from the PRISMA guidelines [30].**

contained “system dynamics” in title, abstract and keywords. The following exclusion criteria were defined:

#### Exclusion Criteria

- The publication was published before 2015.
- The publication type is not a peer-reviewed journal article. For reasons of quality, poster sessions, editorials, interviews, commentaries, conference proceedings, and research-in-progress papers are excluded.
- The publication is considered “off-topic”. This criterion ensures the relevance of the included literature.
- The publication is written in a language other than English.

In the fourth step, the databases used for the SLR were selected and searched. The present study used two databases as well as the journal of the System Dynamics Society: IEEE Xplore Digital Library, ACM Digital Library, and System Dynamics Review.

The search string remained the same for all databases and was applied to search titles, abstracts, and keywords. IEEE Xplore was last accessed on 01/17/2025, ACM on 01/17/2025, and System Dynamics Review on 01/17/2025.

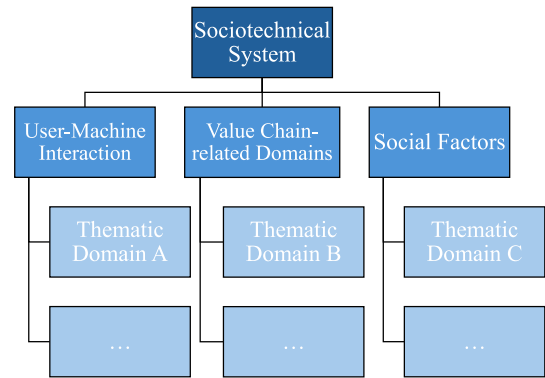


## B. STUDY SELECTION AND EXTRACTION

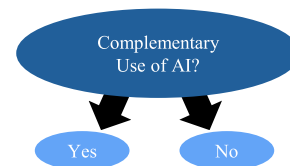
Before screening, 459 articles could be excluded. A total of 194 articles were excluded because they were published before 2015. Furthermore, 184 articles from IEEE Xplore and 21 articles from ACM were excluded due to incorrect article types. Thus, a total of 60 articles were screened. In the screening process, the authors examined the title and abstract for content and semantic relevance to the research question of this paper. To avoid bias in the selection process, all identified publications were individually reviewed by all authors. First, all publications were rechecked for terms contained in the search string. After this first check, the title and abstract were checked using exclusion criteria to assess the semantic context of the respective work. These publications passing this check were downloaded and stored for information extraction. Only publications deemed relevant by all authors were included in the SLR. After review by all authors, the agreement on the results was 98.33% (agreement on 59 of 60 total articles). Based on the discussion, the remaining article was included in the literature review. Automation tools were not used for the search. A total of 19 articles were deemed not relevant to this work. Of the 19 excluded results, 17 results were excluded due to an unsuitable object of investigation, and 2 results were excluded based on their perspective of investigation. After the screening, 41 entries were sought for retrieval. All of the articles could be retrieved, therefore 41 articles were included in the preliminary work and used for the SLR (IEEE Xplore ( $n = 39$ ), ACM ( $n = 1$ ), System Dynamics Review ( $n = 1$ )).

## C. CLASSIFYING THE RESULTS

A systems thinking approach was applied to classify the results and provide a structured framework for assigning the identified use cases to higher-level systems. Bertalanffy's foundational work on systems theory [1] laid the foundation for understanding systems. This understanding of systems is essential for identifying system elements and classifying results into systems. To achieve a systematic classification, this study adopts a sociotechnical systems framework, allowing for the categorization of System Dynamics use cases based on their thematic affiliation. Trist's work [31] on sociotechnical systems provides a theoretical foundation for this concept. Based on this, we derived a framework to structure System Dynamics use cases across different system levels, ensuring a comprehensive and theoretically grounded classification. The framework is presented in Figure 2. The selection of this framework is driven by the broad applicability of System Dynamics, which can be used to analyze technical, social, and value chain-related challenges [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. As part of the SLR, the selected results were reviewed to synthesize thematic domains based on the use cases they presented. The synthesized thematic domains were then assigned to the corresponding system within the framework. This approach ensures methodological



**FIGURE 2.** Graphical representation of the applied sociotechnical systems framework.



**FIGURE 3.** Graphical representation of the used binary AI classification framework.

traceability and contributes to a coherent and structured classification system. In addition to thematic classification, we examined whether artificial intelligence was used complementarily with the System Dynamics methodology. The results were evaluated based on whether AI was used to extend or improve the System Dynamics methodology. The classification employed a binary approach, whereby a rating of “Yes” indicates the complementary use of AI techniques, whereas “No” signifies that no such use was present. This additional perspective provides insights into whether AI is applied to enhance the System Dynamics methodology or to expand its use cases. The process is illustrated in Figure 3.

## IV. RESULTS

This chapter presents the results, which have been systematically classified to address the research questions. First, Tables 3 and 4 provide an overview of the final selection of the relevant literature. This selection has been processed and integrated into the tables to provide a comprehensive overview. Subsequently, a detailed explanation of the identified axes and subsections is provided. This is achieved by explaining the sociotechnical systems, the thematic domains from the use case classification, and the AI classification. Finally, the trend analysis is provided, and the results are summarized in a table that shows the number of publications per year along with the corresponding thematic domains for each year and in total.

### A. SOCIOTECHNICAL SYSTEM CLASSIFICATION

This section contains the system elements of the sociotechnical system. The thematic domains from the use case

**TABLE 3.** Overview of the publications in the literature base. The table presents the authors, the publication year of each study, along with the corresponding sociotechnical system, use case classification and AI classification (Part 1 of 2).

| Author(s)                          | Year | Sociotechnical System       | Use Case Classification   | Use of AI |
|------------------------------------|------|-----------------------------|---------------------------|-----------|
| Zhang et al. [32]                  | 2022 | User-Machine Interaction    | Human-Machine Interaction | No        |
| Li et al. [33]                     | 2020 | Value Chain-related Domains | Energy Management         | No        |
| Peng et al. [34]                   | 2022 | Social Factors              | Urban Transportation      | No        |
| Faezipour and Ferreira [35]        | 2018 | Value Chain-related Domains | Resource Management       | No        |
| Chen et al. [36]                   | 2021 | Value Chain-related Domains | Supply Chain Management   | No        |
| Chavy-Macdonald et al. [37]        | 2019 | Value Chain-related Domains | Product Design            | No        |
| Choi et al. [38]                   | 2022 | Value Chain-related Domains | Economics                 | No        |
| Lu et al. [39]                     | 2023 | Value Chain-related Domains | Resource Management       | No        |
| Liu et al. [40]                    | 2019 | Value Chain-related Domains | Resource Management       | No        |
| Zhang et al. [41]                  | 2021 | Value Chain-related Domains | Product Development       | No        |
| Fenner et al. [42]                 | 2021 | Value Chain-related Domains | IT Management             | No        |
| Liu et al. [43]                    | 2018 | Social Factors              | Environmental Management  | No        |
| Qi et al. [44]                     | 2021 | Social Factors              | Emergency Management      | No        |
| Allahi et al. [45]                 | 2022 | Social Factors              | Emergency Management      | No        |
| Pourbafrani and Van Der Aalst [46] | 2022 | Value Chain-related Domains | Process Optimization      | No        |
| Bezerra et al. [47]                | 2024 | Value Chain-related Domains | IT Management             | No        |
| Restrepo-Tamayo et al. [48]        | 2024 | Social Factors              | Social Influence          | No        |
| Zeng et al. [49]                   | 2021 | Value Chain-related Domains | Innovation Management     | No        |
| Li et al. [50]                     | 2022 | Value Chain-related Domains | Energy Management         | No        |
| Qian et al. [51]                   | 2022 | Value Chain-related Domains | Product Development       | No        |
| Barbalho et al. [52]               | 2022 | Value Chain-related Domains | Product Development       | No        |
| Lu and Fang [53]                   | 2021 | Value Chain-related Domains | Innovation Management     | No        |
| Zarghami and Gunawan [54]          | 2023 | Value Chain-related Domains | Resource Management       | No        |
| Valencia-Calvo et al. [55]         | 2020 | Value Chain-related Domains | Energy Management         | No        |
| Jia et al. [56]                    | 2019 | Social Factors              | Military Tactics          | No        |
| Yung et al. [57]                   | 2023 | Value Chain-related Domains | Innovation Management     | No        |

**TABLE 4.** Overview of the publications in the literature base. The table presents the authors, the publication year of each study, along with the corresponding sociotechnical system, use case classification and AI classification (Part 2 of 2; continued from Table 3).

| Author(s)                    | Year | Sociotechnical System       | Use Case Classification   | Use of AI |
|------------------------------|------|-----------------------------|---------------------------|-----------|
| Appel and Mansouri [58]      | 2022 | Social Factors              | Education                 | No        |
| Jing et al. [59]             | 2022 | Social Factors              | Education                 | No        |
| Khansari et al. [60]         | 2017 | Value Chain-related Domains | Energy Management         | No        |
| Xu et al. [61]               | 2021 | Value Chain-related Domains | Logistics Management      | No        |
| Movahednasab et al. [62]     | 2019 | Value Chain-related Domains | Energy Management         | No        |
| Choudhary and Jain [63]      | 2022 | Value Chain-related Domains | Energy Management         | No        |
| Guo et al. [64]              | 2021 | User-Machine Interaction    | Human-Machine Interaction | No        |
| Faezipour and Faezipour [65] | 2021 | Social Factors              | Medicine                  | No        |
| Xiu et al. [66]              | 2019 | Value Chain-related Domains | Resource Management       | No        |
| Xiang et al. [67]            | 2017 | Value Chain-related Domains | Innovation Management     | No        |
| Diaz et al. [68]             | 2020 | Social Factors              | Emergency Management      | No        |
| Xinfeng et al. [69]          | 2023 | Value Chain-related Domains | Risk Management           | No        |
| Xiu and Zhao [70]            | 2021 | Social Factors              | Environmental Management  | Yes       |
| Luna-Reyes and Harrison [71] | 2024 | Value Chain-related Domains | Innovation Management     | No        |
| Green et al. [72]            | 2024 | Social Factors              | Emergency Management      | No        |

classification that could be synthesized from the literature were assigned to these system elements.

#### 1) SYSTEM 1 - USER-MACHINE INTERACTION

This system incorporates all thematic domains constituting the User-Machine Interaction system. A constituent element of this system is the thematic domain of Human-Machine Interaction (HMI).

##### *a: HUMAN-MACHINE INTERACTION*

This subsection includes publications that explore use cases in the field of HMI, all of which incorporate a System Dynamics model applied to a problem-specific context. These models are designed to optimize HMI by leveraging System Dynamics in real-world applications. Zhang et al. [32] introduce a game-based model integrated with System Dynamics to enhance HMI in intelligent aircraft cockpits, conducting a sensitivity analysis on task load dynamics and decision-making performance. Similarly, Guo et al. [64] combine System Dynamics with coupling theory to assess

the efficiency of human-machine systems in plateau mining operations, offering actionable insights into optimizing work-rest schedules and improving HMI under extreme conditions.

#### 2) SYSTEM 2 - VALUE CHAIN-RELATED DOMAINS

This system incorporates all thematic domains constituting the Value Chain-related Domains system. The aforementioned system encompasses the thematic domains Energy Management, Resource Management, Supply Chain Management, Product Design, Economics, Product Development, IT Management, Process Optimization, Innovation Management, Logistics Management, and Risk Management.

##### *a: ENERGY MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Energy Management. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to optimize existing energy markets, structures, and systems through



System Dynamics, informed by practical applications and real-world use cases. Li et al. [33] integrate System Dynamics with game theory to refine government subsidy strategies for Shore Power (SP) technology, addressing challenges such as information asymmetry, decision-making cycles, and technological advancements while offering insights into subsidy efficiency and SP reliability. Similarly, Li et al. [50] present an integrated methodology that combines System Dynamics with the ImPACT equation to project long-term electricity consumption, incorporating emerging factors such as renewable energy adoption, technological progress, and environmental policies to support strategic energy system planning. Valencia-Calvo et al. [55] introduce a hybrid approach that merges System Dynamics with differential equations to examine non-smooth dynamics in electricity markets, identifying critical thresholds and evaluating the broader implications of investment decisions on market stability. Khansari et al. [60] develop a comprehensive System Dynamics model to analyze the interplay between social progress, energy consumption behavior, and CO<sub>2</sub> emissions, emphasizing the pivotal role of government interventions and information dissemination in promoting sustainable energy practices. Movahednasab et al. [62] employ System Dynamics in conjunction with Monte Carlo simulations to assess long-term scenarios for natural gas and electricity exports, identifying forward contracts as the most economically advantageous strategy for energy-exporting nations. Lastly, Choudhary and Jain [63] construct an integrated System Dynamics model to examine supply chain disruptions in the oil and gas sector during the COVID-19 pandemic, shedding light on key resilience mechanisms and the interdependencies among economic, environmental, and social sustainability dimensions.

#### *b: RESOURCE MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Resource Management. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to improve decision-making processes, optimize resource allocation, and enhance sustainability, drawing on practical applications and real-world use cases. Faezipour and Ferreira [35] develop a System Dynamics model to assess water sustainability in hospitals by incorporating social, economic, and environmental factors. Their approach provides a simulation tool designed to optimize water usage, lower costs, and refine sustainability strategies. A different focus is taken by Lu et al. [39], who propose a three-dimensional mechanism integrating System Dynamics to enhance the allocation of regional science and technology resources. Their model improves sharing efficiency and value generation through coordinated government-market dynamics and benefit-cost assessments. Another perspective is provided by Liu et al. [40], who introduce a hybrid approach combining Real Options Analysis with System Dynamics to improve investment decision-making in mining

projects. Their study addresses dynamic uncertainties and managerial flexibilities, demonstrated through the valuation of a uranium mine in China. Meanwhile, Zarghami and Gunawan [54] develop a System Dynamics model to project the impact of population growth on water distribution network robustness. Their findings highlight the crucial role of even population distribution in strengthening system resilience. Lastly, Xiu et al. [66] construct a holistic System Dynamics model that integrates technical and economic parameters to support sustainable operations in metal mining. Their analysis offers actionable insights through scenario and uncertainty assessments, applied to the Sanshandao gold mine.

#### *c: SUPPLY CHAIN MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Supply Chain Management. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to enhance capacity planning by drawing on practical applications and real-world use cases. Chen et al. [36] develop a System Dynamics model to improve capacity planning in component supply for complex product manufacturing. Their study highlights the importance of information sharing and the integration of backup suppliers in reducing supply chain disruptions.

#### *d: PRODUCT DESIGN*

This category subsection encompasses literature with a use case in the field of Product Design. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to identify and optimize dynamic design priorities, drawing on practical applications and real-world use cases. Chavy-Macdonald et al. [37] propose a multimethodological approach that integrates System Dynamics with scenario planning to determine dynamic design priorities for product architectures. Their study explores the relationship between uncertainty recognition and concept generation, aiming to enhance product changeability.

#### *e: ECONOMICS*

This category subsection encompasses literature with a use case in the field of Economics. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models seek to analyze socioeconomic impacts and other general economic subjects, including factors such as efficiency and competitiveness, informed by practical applications and real-world use cases. Choi et al. [38] develop a hybrid System Dynamics and Analytic Hierarchy Process (AHP) model to quantify the long-term socioeconomic ripple effects of integrating national construction standards, enhancing construction efficiency, international competitiveness, and safety.

#### *f: PRODUCT DEVELOPMENT*

This category subsection encompasses literature with a use case in the field of Product Development. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to optimize the product development process and its associated factors, drawing on practical applications and real-world use cases. Zhang et al. [41] develop a System Dynamics model to quantify the impact of requirement changes on product development progress. Their study highlights the critical role of timing, change probability, and rework dynamics. Qian et al. [51] address another aspect of product development. Their System Dynamics model examines the interaction between technology management and technological capability, identifying phase-specific strategies to enhance new product development through tailored resource alignment. Barbalho et al. [52] take a different approach with a predictive System Dynamics model. Their research focuses on the influence of product complexity and team seniority on project performance, enabling more precise forecasting of project schedules in complex new product development projects.

#### *g: IT MANAGEMENT*

This category subsection encompasses literature with a use case in the field of IT Management. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to improve and optimize strategic decision-making processes and capacity management, drawing on practical applications and real-world use cases. Fenner et al. [42] introduce a business-driven simulation model that combines System Dynamics, Balanced Scorecard, and AHP to enhance capacity management for Infrastructure as a Service (IaaS) providers. Their approach aligns technical capabilities with business objectives to improve efficiency and customer satisfaction. Bezerra et al. [47] take a different perspective by developing CapSim, a System Dynamics model designed to support IT sourcing management. Their study integrates strategic goals with operational decisions and is validated through long-term case studies, facilitating cost efficiency and optimized resource allocation.

#### *h: PROCESS OPTIMIZATION*

This category subsection encompasses literature with a use case in the field of Process Optimization. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to optimize processes by uncovering dynamic interrelationships, informed by practical applications and real-world use cases. Pourbafrani and Van der Aalst [46] propose a data-driven framework combining Process Mining and System Dynamics to automatically generate simulation models, enabling scenario-based analysis and process optimization through

the discovery of dynamic interrelationships in business processes.

#### *i: INNOVATION MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Innovation Management. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to analyze innovation ecosystems, networks, and dynamics, drawing on practical applications and real-world use cases. Zeng et al. [49] develop a System Dynamics model to simulate and forecast regional technology innovation dynamics. Their study integrates Foreign Direct Investment (FDI), Intellectual Property Rights (IPR) policies, and Research and Development (R&D) investments, providing actionable insights for policymakers to optimize innovation strategies. Focusing on another aspect of innovation, Lu and Fang [53] examine the performance of collaborative innovation networks. Their System Dynamics model highlights the role of government investment, technology intermediaries, and enterprise innovation in enhancing network efficiency and economic outcomes. A different perspective is offered by Yung et al. [57], who investigate interactions within high-tech innovation ecosystems. Using two aerospace instrument case studies, their model explores how collaboration between government, universities, and enterprises influences innovation performance and system resilience. Shifting attention to technological adoption, Xiang et al. [67] employ a System Dynamics model to simulate the scale evolution of Electric Vehicles (EV). Their research integrates social, economic, and technological factors to identify effective policies and strategies for EV adoption. Expanding the scope to AI-driven transformation, Luna-Reyes and Harrison [71] introduce a System Dynamics model to evaluate enterprise-wide strategies for AI capability development in public administration. Their findings emphasize the importance of IT governance, human resource development, and data interoperability in fostering AI integration.

#### *j: LOGISTICS MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Logistics Management. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to enhance logistics systems, drawing on practical applications and real-world use cases. Xu et al. [61] develop a System Dynamics model to assess governance measures for container port congestion. Their study evaluates the effectiveness of infrastructure expansion, multimodal transport, smart strategies, and interconnectivity, providing insights based on a case study of Shanghai Port.

#### *k: RISK MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Risk Management. The categorized publications each include a System Dynamics model

applied to problem-related topics. These models aim to identify strategic risks, drawing on practical applications and real-world use cases. Xinfeng et al. [69] develop a System Dynamics-based model to assess strategic risks in equipment system development. Their study examines causal feedback loops and proposes measures to enhance risk management and improve strategic decision-making.

### 3) SYSTEM 3 - SOCIAL FACTORS

This system incorporates all thematic domains constituting the Social Factors system. The aforementioned domains encompass Urban Transportation, Environmental Management, Emergency Management, Social Influence, Military Tactics, Education, and Medicine.

#### *a: URBAN TRANSPORTATION*

This category subsection encompasses literature with a use case in the field of Urban Transportation. The categorized publications each include a System Dynamics model applied to a problem-related topic. These models aim to enhance urban transportation systems through System Dynamics, drawing on practical applications and real-world use cases. Peng et al. [34] develop a System Dynamics model to optimize passenger flows in metro station entrance halls. Their study investigates bottleneck regions and determines optimal aisle lengths, improving efficiency without requiring additional infrastructure.

#### *b: ENVIRONMENTAL MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Environmental Management. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to analyze sustainable development strategies and assess the diffusion of alternative technologies, informed by practical applications and real-world use cases. Liu et al. [43] develop a System Dynamics model to examine the diffusion of alternative powertrain technologies in the U.S. market. Their study integrates consumer behavior, government incentives, and manufacturer strategies to evaluate long-term adoption rates and emission reductions. In a different context, Xiu and Zhao [70] combine System Dynamics with an optimized K-means algorithm to analyze the sustainable development of port economies. Their model focuses on population, environment, resources, and economic subsystems to identify actionable strategies for sustainability and competitiveness.

#### *c: EMERGENCY MANAGEMENT*

This category subsection encompasses literature with a use case in the field of Emergency Management. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to evaluate humanitarian interventions and optimize crisis management strategies, informed by practical applications and real-world use cases. Qi et al. [44] develop a System Dynamics model to optimize emergency supply management. Their study

analyzes the interplay of government reserves, production, imports, and demand under both peacetime and wartime scenarios, highlighting the efficiency of comprehensive control strategies. Taking a different approach, Allahi et al. [45] present a System Dynamics model to assess the impact of cash-based interventions on refugee dignity. Their research underscores the need to balance short-term relief efforts with long-term infrastructure investments for sustainable humanitarian aid. Focusing on disaster recovery, Diaz et al. [68] develop a System Dynamics model to simulate housing reconstruction after disasters. Their model integrates material flows, workforce dynamics, and demand to provide actionable insights for humanitarian logistics and disaster management. Expanding the scope to pandemic preparedness, Green et al. [72] utilize System Dynamics to integrate epidemiological and hospital-resource models. Their study offers scenario-based insights to optimize interventions and resource planning.

#### *d: SOCIAL INFLUENCE*

This category subsection encompasses literature with a use case in the field of Social Influence. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to integrate social factors into productivity models, drawing on practical applications and real-world use cases. Restrepo-Tamayo et al. [48] propose a methodology that combines System Dynamics, systematic mapping, and survey-based studies. Their approach incorporates social factors such as communication, leadership, and teamwork into productivity models for software development teams, addressing existing gaps in the analysis of non-technical influences.

#### *e: MILITARY TACTICS*

This category subsection encompasses literature with a use case in the field of military tactics. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to assess the operational effectiveness of military-related innovations and strategies, drawing on practical applications and real-world use cases. Jia et al. [56] develop a System Dynamics model to evaluate the combat effectiveness of swarming Uncrewed Aerial Vehicles (UAVs). Their study highlights the tactical advantages of such systems in military operations, demonstrating their potential to outperform traditional high-value weapon platforms in combat scenarios.

#### *f: EDUCATION*

This category subsection encompasses literature with a use case in the field of Education. The categorized publications each include a System Dynamics model applied to problem-related topics. These models aim to evaluate educational reforms and optimize education processes, drawing on practical applications and real-world use cases. Appel and Mansouri [58] develop a validated System Dynamics model

to examine STEM education and career pipelines. Their study highlights mentorship and peer programs as key interventions for reducing attrition while enhancing student resilience and career success. Shifting the focus to technological education, Jing et al. [59] present a System Dynamics model that analyzes factors influencing AI talent cultivation. Their research identifies essential educational reforms and technological integration strategies designed to bridge the gap between educational supply and industry demands.

g: MEDICINE

This category subsection encompasses literature with a use case in the field of Medicine. The categorized publications each include a System Dynamics model applied to problem-related topics. These models are designed to assess existing and emerging medical practices and challenges, drawing on practical applications and real-world use cases. Faezipour and Faezipour [65] apply System Dynamics to examine the effectiveness of smartphone-based Electrocardiogram (ECG) monitoring systems. Their study explores the interplay between system design, user satisfaction, and cost dynamics, emphasizing their role in optimizing service delivery.

B. AI CLASSIFICATION

The purpose of this section is to determine whether artificial intelligence has been used to complement or improve the System Dynamics method or its models.

1) NO - AI HAS NOT BEEN USED

This category comprises publications that did not employ AI to complement or improve System Dynamics methodology. This category encompasses 40 out of the total of 41 papers.

2) YES - AI HAS BEEN USED

This category includes publications that used AI in their work to complement or improve the System Dynamics methodology. This criterion was met by one out of 41 papers. Xiu and Zhao [70] supplemented the System Dynamics model with an optimized K-means algorithm that performed cluster analyses for pattern recognition and at the same time accelerated the calculation of central model parameters. This combination enabled a more efficient simulation of the interactions between population, environmental, resource, and economic indicators in order to derive sound strategic decisions for a sustainable port economy.

C. TREND ANALYSIS

In this section, we structure the synthesized literature to identify research trends. The number of relevant publications was structured by year. This is depicted in Figure 4. In addition to categorizing the relevant publications by year, Figure 5 presents a classification of the publications into their respective thematic domains. Furthermore, it illustrates the number of publications within each thematic domain for each respective year. Additionally, Figure 5 provides a comprehensive overview of the total number of publications

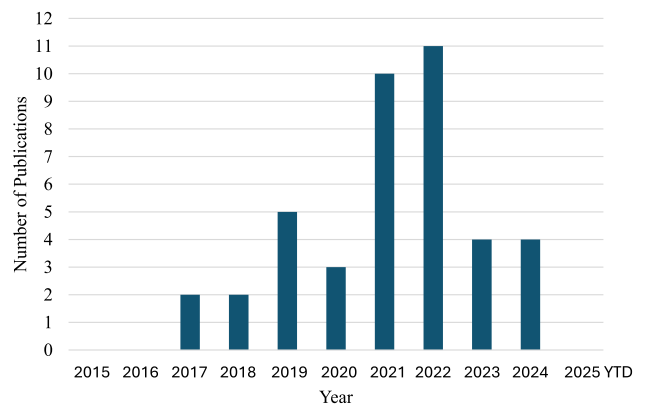


FIGURE 4. Graphical illustration of the number of publications in the literature base structured by year.

| Domain \ Year        | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | Σ |
|----------------------|------|------|------|------|------|------|------|------|---|
| HMI                  |      |      |      |      | 1    | 1    |      |      | 2 |
| Energy Mgmt.         | 1    |      | 1    | 2    |      | 2    |      |      | 6 |
| Urban Transpor.      |      |      |      |      |      | 1    |      |      | 1 |
| Resource Mgmt.       |      | 1    | 2    |      |      |      | 2    |      | 5 |
| Supply Chain Mgmt.   |      |      |      |      | 1    |      |      |      | 1 |
| Product Design       |      |      | 1    |      |      |      |      |      | 1 |
| Economics            |      |      |      |      |      | 1    |      |      | 1 |
| Product Dev.         |      |      |      |      | 1    | 2    |      |      | 3 |
| IT Mgmt.             |      |      |      |      | 1    |      |      | 1    | 2 |
| Environmental Mgmt.  |      | 1    |      |      | 1    |      |      |      | 2 |
| Emergency Mgmt.      |      |      |      | 1    | 1    | 1    |      | 1    | 4 |
| Process Optimization |      |      |      |      |      | 1    |      |      | 1 |
| Social Influence     |      |      |      |      |      |      |      | 1    | 1 |
| Innovation Mgmt.     | 1    |      |      |      | 2    |      | 1    | 1    | 5 |
| Military Tactics     |      |      | 1    |      |      |      |      |      | 1 |
| Education            |      |      |      |      |      | 2    |      |      | 2 |
| Logistics Mgmt.      |      |      |      |      | 1    |      |      |      | 1 |
| Medicine             |      |      |      |      | 1    |      |      |      | 1 |
| Risk Mgmt.           |      |      |      |      |      |      | 1    |      | 1 |

FIGURE 5. Graphical table of the number of publications in each thematic domain per year.

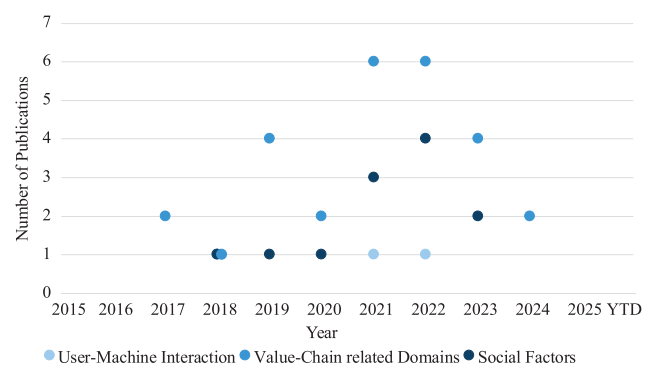


FIGURE 6. Graphical illustration of the number of publications in each system per year.

in each thematic domain. Figure 6 shows the number of publications per year in each system.

V. DISCUSSION

The first research question aimed to identify and classify the System Dynamics use cases. As part of the Systematic Literature Review, 19 thematic domains were extracted and subsequently classified into three overarching systems using a sociotechnical framework: (1) User-Machine Interaction, (2) Value Chain-related Domains, and (3) Social Factors. The



application of this framework enabled a structured categorization of the use cases. It became evident that the number of studies varied significantly across the three systems. The following discussion explores the contexts in which System Dynamics has become particularly well-established and what academic or methodological factors might explain the lower representation in other areas.

With 27 of the 41 identified publications, Value Chain-related Domains constitute the most represented system. This can be attributed to the high practical relevance of System Dynamics in economically oriented contexts, where process optimization, resource allocation, or strategic decision support are central concerns. System Dynamics allows for the simulation of long-term effects, systemic feedback, and uncertainty. These features are especially valuable to decision-makers in business and policy environments. Several studies also demonstrate a trend toward hybrid modeling approaches, combining System Dynamics with methods such as Monte Carlo simulation, Analytic Hierarchy Process (AHP), or process mining. This highlights the methodological flexibility and practical value of System Dynamics in data-intensive, real-world scenarios.

The Social Factors system comprises 12 studies covering areas such as emergency management, education, and public health. These fields are of high societal relevance and often shaped by complex stakeholder environments. Although System Dynamics has been applied successfully in some cases to assess policy interventions or resource planning, the overall number of studies is comparatively low. A possible explanation lies in the limited economic monetization potential of these domains. While System Dynamics is frequently used in corporate contexts to improve efficiency or achieve strategic advantages, application in social systems such as education, public administration, or healthcare often lacks immediate financial incentives. As a result, fewer research efforts may have been directed toward developing and applying System Dynamics models in these areas, despite their high societal impact.

Only two publications were assigned to the User-Machine Interaction system. Both studies address highly technical scenarios, such as intelligent cockpit design or automation in mining. They combine System Dynamics with game-theoretical or coupling-based approaches. This limited number may be due to a methodological mismatch: System Dynamics operates at an aggregated systems level and is geared toward delayed feedback processes, while human-machine systems often require fine-grained, adaptive, and real-time modeling. These requirements could be better met by agent-based models, cognitive simulations, or machine learning techniques.

It is also conceivable that use cases in User-Machine Interaction are inherently difficult to translate into traditional System Dynamics models. These scenarios frequently involve situational decision-making, individual interactions, and cognitive dynamics that do not easily lend themselves to aggregation into feedback-based system structures. The

comparatively high level of abstraction in these systems, along with their low structural repeatability, may further complicate the application of System Dynamics in this context.

Overall, the distribution of System Dynamics use cases highlights both the strengths and the methodological boundaries of the approach. System Dynamics is predominantly applied in areas where aggregated system structures with long-term effects are modeled. It is less frequently used in domains where individual decision-making, decentralization, or rapid responsiveness are central. However, this does not imply that System Dynamics is unsuitable for these areas. Rather, disciplinary traditions and methodological perceptions may influence its adoption. A more nuanced examination of these dynamics could guide future research and extend the methodological reach of System Dynamics through hybrid or interdisciplinary approaches.

Compared to previous reviews, particularly the work of Zanker et al. [4], which structured System Dynamics applications into thematic categories, this study offers a complementary perspective through a systems thinking lens. By classifying 19 thematic domains into sociotechnical systems, it provides a systemic understanding of application contexts. Notably, this classification explicitly recognizes User-Machine Interaction as a distinct and emerging domain, which has not been singled out in previous overviews, but has already been addressed using the System Dynamics approach [73]. As human users increasingly interact with machines through artificial intelligence in real-time and decision-making settings, dynamic modeling of these interactions becomes more relevant, positioning User-Machine Interaction as a promising and timely area for future System Dynamics use cases.

The second research question was to investigate the extent to which AI techniques, in particular LLMs, were applied in a complementary manner with the System Dynamics methodology in the identified use cases. The results of this study show that only one out of the 41 analyzed publications used AI as a complementary approach to enhance the System Dynamics methodology. This limited adoption may be attributed to two key factors.

First, LLMs such as ChatGPT only became widely accessible at the end of 2022. Prior to that, researchers had limited opportunities to experiment with AI-supported enhancements to System Dynamics. While other AI technologies had already been available earlier, methodological and technical barriers may have hindered their application.

Second, specific integration methods that enable a connection between System Dynamics models and LLMs, such as ChatPySD [27], were only developed and published in 2025. The absence of established methods for AI-System Dynamics integration during the review period likely contributed to the limited adoption as well.

These findings suggest that AI has so far played only a minor role in System Dynamics use cases. This limited use is likely due less to conceptual incompatibility and more to



restricted accessibility and the lack of technical standards for integration. The late availability of powerful AI technologies and the absence of established interfaces may have prevented the potential synergies between AI and System Dynamics from being fully realized thus far.

Given that AI techniques are already being used in data analysis [74], simulation modeling [75], and decision support [76], it may only be a matter of time before AI is more extensively integrated into the System Dynamics methodology. This expectation is supported by recent studies demonstrating how LLMs can generate causal loop diagrams from textual data using natural language processing [28], [29]. Furthermore, the ChatPySD method [27] enables embedding, executing, and interacting with System Dynamics models within ChatGPT through natural language interaction. This AI integration has the potential to complement traditional SD modeling by improving model transparency, accessibility, and interpretability, while also enabling HMI [27]. In sum, AI-driven enhancements could help mitigate some of the current limitations of System Dynamics and expand its applicability.

The third research question was used to investigate what trends in publishing in System Dynamics research can be observed over time. The analysis shows that no relevant publications were identified for the years 2015 and 2016. This may be due either to a generally lower volume of System Dynamics research during that period or to the exclusion criteria of the review, which filtered out studies not meeting specific methodological or topical standards.

From 2017 to 2019, the number of relevant publications increased moderately, followed by a notable surge in 2020. The years 2021 and 2022 represented the peak of identified research activity, with 11 and 10 publications, respectively. A likely explanation is the growing methodological recognition of System Dynamics as a tool for strategic decision support, coupled with a broader trend toward modeling-based research. In addition, pandemic-related lockdowns may have indirectly contributed to this increase: as field and lab studies became more difficult, academic focus shifted toward theoretical and simulation-based approaches, potentially boosting publication output in this domain.

In 2023 and 2024, publication numbers declined to four per year. Whether this marks a temporary fluctuation or the onset of a longer-term trend remains uncertain. Possible explanations include topic saturation or a shift in methodological preferences. At the same time, the rising availability of AI-enhanced tools may open new avenues for System Dynamics applications, suggesting that future developments could differ substantially from previous patterns. No relevant publications were identified for the year 2025 within the scope of this SLR. This is likely attributable to the review period ending on January 17, 2025, with potentially relevant studies being published after this date. A more detailed view of the three classified application systems reveals distinct temporal dynamics:

Value Chain-related Domains have consistently dominated the publication landscape. Growth in this area began early and peaked in 2021 with eight studies. This trend reflects the strong alignment of System Dynamics with economically driven contexts, where process optimization, resource allocation, and strategic planning are key concerns. Furthermore, this system demonstrates a notable openness to hybrid modeling approaches, such as Monte Carlo simulations and AHP, enhancing the method's practical utility in data-rich environments.

Social Factors showed a moderate increase in research activity between 2017 and 2022, followed by a visible decline. The initial growth may have been driven by heightened attention to global challenges such as climate change and crisis management, which require systemic perspectives. The subsequent drop could indicate reduced perceived urgency or a methodological shift in addressing these topics. Additionally, many social issues are inherently difficult to formalize, limiting their compatibility with System Dynamics. The lack of financial incentives and established modeling traditions may further restrict its uptake in this domain. It is also conceivable that the comparatively steep learning curve of System Dynamics hinders adoption in fields where quantitative modeling is not a standard practice.

User-Machine Interaction remains the least represented application system, with only two identified publications, both from 2021 and 2022. This suggests that it is still an emerging area within the field. Due to the limited data, it is currently not possible to draw robust conclusions about long-term trends or future research potential.

## A. IMPLICATIONS

The application of System Dynamics could be of interest to companies, as it can be used to optimize various operational processes. Potential use cases include the 19 thematic domains identified in this SLR. As a result, financial expenditures could be reduced, and competitive advantages could be achieved, particularly over companies that do not apply the System Dynamics approach. System Dynamics can also serve as a methodological approach to strengthening the resilience of supply chains against disruptions and crises. Similarly, System Dynamics could be valuable for medical professionals in evaluating treatment methods by identifying systemic interdependencies and feedback loops, ultimately enhancing treatment effectiveness. Furthermore, the application of System Dynamics in the political domain could be highly relevant, as it allows for the evaluation of different policy measures and facilitates data-driven decision-making. The underlying models could be published, enhancing transparency and making the decision-making process more comprehensible for external stakeholders.

## VI. CONCLUSION

This study conducted a Systematic Literature Review to identify and classify the use cases of System Dynamics across all thematic domains. To achieve this, thematic domains

were synthesized from the literature and systematically classified within a sociotechnical framework. Additionally, the study examined the extent to which artificial intelligence has complemented System Dynamics methodologies and identified trends in System Dynamics use cases over time. Most studies focused on Value Chain-related Domains (27), followed by Social Factors (12) and User—Machine Interaction (2). A key insight is the limited integration of AI into System Dynamics applications. Despite AI's increasing role in computational modeling, only one study employed an AI-based method, using a K-means clustering algorithm to enhance a System Dynamics model. The analysis also reveals a peak in research activity in 2021 and 2022, followed by a decline from 2023 onward. Whether this marks a temporary fluctuation or a shift in methodological focus remains to be seen. This study makes two key contributions. First, it provides a comprehensive and up-to-date overview of System Dynamics use cases, systematically categorizing them within a sociotechnical framework. Second, it identifies a significant research gap in AI-enhanced System Dynamics modeling, emphasizing the lack of established methodologies for AI-driven System Dynamics approaches. These findings underscore the need for further exploration and development of AI-driven System Dynamics methodologies.

### A. LIMITATIONS

We decided to consider only journal articles for this Systematic Literature Review. As a result, relevant publications may have been excluded due to this selection criterion. Furthermore, we chose a 10-year observation period to include only recent publications. Consequently, relevant studies published outside this timeframe may have been overlooked.

### B. FUTURE RESEARCH

Several directions for future research can be identified. These include both methodological expansions of this study and investigations into the complementary integration of AI with System Dynamics. One potential avenue for future research is the complementary integration of AI approaches into existing or newly developed System Dynamics models. The current body of literature indicates that only one study has combined AI with System Dynamics, highlighting a significant research gap. Future studies could explore how AI-driven approaches, such as ChatPySD [27], can be used to optimize and enhance System Dynamics models. Furthermore, there is a need to explore new methodologies for the complementary use of AI to advance the System Dynamics approach. This could include the development of novel methods that improve accessibility, validation, transparency, or other aspects of System Dynamics models.

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